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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

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TITLE OF THE PROJECT	DOSETWIN AI: A DIGITAL TWIN FRAMEWORK FOR PERSONALIZED DRUG RESPONSE SIMULATION AND MEDICATION CONFLICT PREVENTION	
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Abstract:

DoseTwin AI is a Digital Twin-based healthcare framework designed to support personalized drug-response simulation and medication conflict prevention. In conventional healthcare systems, medication decisions are often based on generalized clinical information and predefined drug-interaction databases, which may not fully consider the unique characteristics of an individual patient. Differences in medical history, existing medications, laboratory results, demographic information, and other patient-specific factors can influence how a person responds to a particular drug. Therefore, there is a need for an intelligent system that can combine patient-specific information with medication knowledge to support safer and more personalized treatment decisions.

The proposed DoseTwin AI framework addresses this challenge by creating a virtual Digital Twin of the patient. The Digital Twin maintains a structured representation of relevant patient information, including medical records, medication history, laboratory reports, demographic details, and clinical parameters. This virtual representation enables the system to analyze and simulate possible medication responses based on the patient's individual profile rather than relying only on generalized information.

To ensure standardized healthcare data handling, HL7 FHIR is incorporated into the framework for representing and exchanging healthcare information in a structured and interoperable manner. FastAPI-based microservices provide the backend architecture and enable different system components to communicate through well-defined APIs. This microservices approach allows the major functions of the framework, including patient data processing, Digital Twin management, drug analysis, and conflict detection, to operate as coordinated services.

The intelligent analysis component uses TensorFlow to support machine-learning-based drug-response prediction. Patient-specific information can be processed to identify patterns and estimate possible responses to selected medications. In parallel, a Knowledge Graph represents relationships between drugs, diseases, medications, and potential interactions. By combining the predictive capability of TensorFlow with the relationship-based reasoning of the Knowledge Graph, the framework can analyze medication combinations and identify potential conflicts that may require attention.

The framework performs drug-response simulation and medication conflict detection using the information available in the patient's Digital Twin. The simulated results can be used to understand possible medication effects and identify potential risks before a treatment decision is made. PostgreSQL is used to store structured patient, medication, interaction, and simulation-related information, providing persistent and organized data management. Docker is used for containerizing the services, supporting consistent execution, deployment, and scalability of the application.

The overall workflow of DoseTwin AI consists of collecting patient and medication information, standardizing the data through HL7 FHIR, processing it through FastAPI microservices, creating and updating the Digital Twin, performing AI-based drug-response analysis, evaluating medication relationships through the Knowledge Graph, and generating personalized analysis results. The system can provide personalized drug-response information, potential medication conflict alerts, risk analysis, and supporting treatment insights.

Thus, DoseTwin AI combines Digital Twins, HL7 FHIR, TensorFlow, Knowledge Graphs, FastAPI, PostgreSQL, and Docker into a unified framework for personalized medication analysis. The proposed approach aims to demonstrate how Digital Twin technology and microservices can be applied to healthcare to improve medication safety, support individualized treatment analysis, reduce potential medication-related risks, and contribute to better treatment outcomes.